

Week 2 • Day 4

Random Forest Classifier

Trained a 200-tree Random Forest ensemble and extracted feature importance rankings

DATE	INTERN NAME	PROJECT	MENTOR
Day 11 of 28 Week 2		Customer Churn Prediction & LTV Engine	

■ Technologies Used Today

Python

Scikit-Learn

Pandas

Matplotlib

joblib

NumPy

■ Today's Key Metrics

200

Trees Trained

>0.82

AUC-ROC Target

20

Features Ranked

3

Models in Pipeline

■ Day Objectives

- Understand decision trees and Random Forest ensemble method
- Train RandomForestClassifier with 200 trees
- Extract and plot top-20 feature importances
- Compare AUC with Logistic Regression baseline
- Save models/random_forest.pkl

■ Tasks Completed Today

1. Created src/models/random_forest.py
2. Loaded data from telco_features.csv and called preprocess_for_ml()
3. Confirmed: Random Forest does NOT need feature scaling (tree-based algorithm)

4. Trained RandomForestClassifier(n_estimators=200, class_weight='balanced', random_state=42, n_jobs=-1)
5. Generated predictions and probabilities on test set
6. Calculated AUC-ROC — improved over Logistic Regression
7. Printed full classification report
8. Extracted model.feature_importances_ — importance score for each feature
9. Created horizontal bar chart of top-20 features by importance
10. Saved reports/figures/11_rf_feature_importance.png
11. Saved models/random_forest.pkl
12. Committed to GitHub

■ Concepts Learned

- Decision tree: splits data by threshold questions on features to reach a leaf node prediction
- Random Forest: trains N trees on random row/feature subsets, majority-votes the result
- Bagging (Bootstrap Aggregating): why training on random subsets reduces variance
- Feature importance in RF: how much each feature reduces impurity across all trees
- n_jobs=-1: uses all CPU cores in parallel for faster training
- Why tree-based models don't need scaling: they split on thresholds, not magnitudes
- Ensemble advantage: 200 trees is more accurate and robust than 1 tree

■ Tools / Software Used

Tool / Library	Version / Notes	Status
RandomForestClassifier	scikit-learn 1.3.2	✓ Trained with 200 trees

■ Outputs / Deliverables Produced

- ◆ src/models/random_forest.py — training and evaluation script
- ◆ models/random_forest.pkl — saved 200-tree Random Forest
- ◆ reports/figures/11_rf_feature_importance.png — top 20 features chart
- ◆ AUC-ROC score (higher than LR baseline)
- ◆ Feature ranking: tenure, monthlycharges, totalcharges top 3
- ◆ GitHub commit with model and importance chart

■ Problems Encountered & Solutions

Problem Faced	How I Solved It
Training took longer than expected	Added n_jobs=-1 to use all CPU cores — training time reduced significantly

■ GitHub Commit Record

Commit Message	Week2-Day4: Random Forest — 200 trees trained, feature importance chart, AUC documented
Branch	main
Files Changed	src/models/random_forest.py, models/random_forest.pkl, reports/figures/11_rf_feature_importance.png
Commit Hash	(fill after push)

■ End-of-Day Submission Checklist

■ RF model trained (200 trees)	■ AUC higher than LR
■ Feature importance extracted	■ Top-20 features chart saved
■ Tenure in top-3 features	■ random_forest.pkl saved
■ Chart 11 saved	■ n_jobs=-1 used
■ Classification report printed	■ Day 11 commit on GitHub

■ Self Assessment

9/10 Self Rating	5 Hours Spent	Week 2, Day 4 Progress	11/28 Day in Program
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■ Key Learnings & Reflections

- ★ Tenure and monthly charges dominate feature importance — consistent with EDA findings
- ★ Random Forest outperforms Logistic Regression because it captures non-linear interactions
- ★ n_jobs=-1 is always worth adding for tree models — free speed improvement
- ★ Feature importance from RF confirms which features are genuinely predictive

■ Plan for Tomorrow

- Train XGBoost with gradient boosting
- Set scale_pos_weight for class imbalance

- Compare all three model ROC curves on one plot
- Select the best model for production

INTERN SIGNATURE	MENTOR SIGNATURE	DATE SUBMITTED